

- 3 (a) acceleration proportional to displacement/distance from fixed point
and in opposite directions/directed towards fixed point
- M1
A1 [2]

(b) energy = $\frac{1}{2}m\omega^2x_0^2$ and $\omega = 2\pi f$
 $= \frac{1}{2} \times 5.8 \times 10^{-3} \times (2\pi \times 4.5)^2 \times (3.0 \times 10^{-3})^2$
 $= 2.1 \times 10^{-5} \text{ J}$

C1
C1
A1 [3]

- (c) (i) at maximum displacement
above rest position
- M1
A1 [2]

(ii) acceleration = $(-)\omega^2x_0$ and acceleration = 9.81 or g
 $9.81 = (2\pi \times 4.5)^2 \times x_0$
 $x_0 = 1.2 \times 10^{-2} \text{ m}$

C1
A1 [2]

- 4 (a) e.g. storing energy